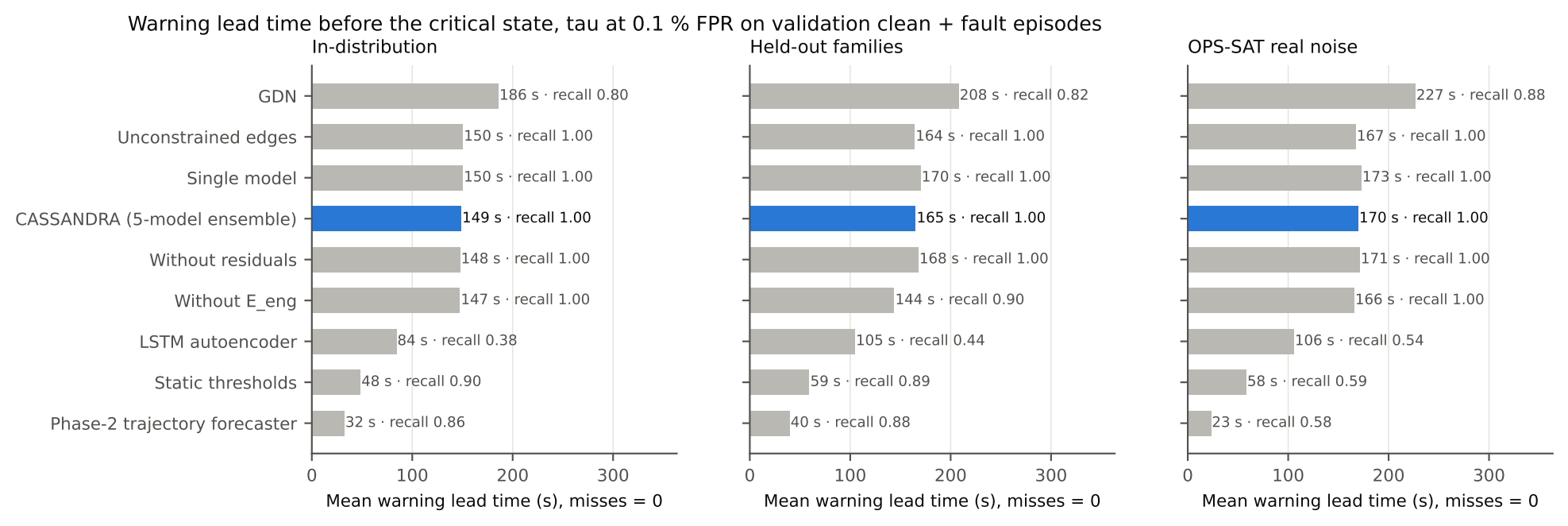
Warning lead time before the critical state, tau at 0.1 % FPR on validation clean + fault episodes
In-distribution
Held-out families
OPS-SAT real noise
GDN
Unconstrained edges
Single model
CASSANDRA (5-model ensemble)
Without residuals
Without E_eng
LSTM autoencoder
Static thresholds
Phase-2 trajectory forecaster
186 s · recall 0.80
150 s · recall 1.00
150 s · recall 1.00
149 s · recall 1.00
148 s · recall 1.00
147 s · recall 1.00
84 s · recall 0.38
48 s · recall 0.90
32 s · recall 0.86
208 s · recall 0.82
164 s · recall 1.00
170 s · recall 1.00
165 s · recall 1.00
168 s · recall 1.00
144 s · recall 0.90
105 s · recall 0.44
59 s · recall 0.89
40 s · recall 0.88
227 s · recall 0.88
167 s · recall 1.00
173 s · recall 1.00
170 s · recall 1.00
171 s · recall 1.00
166 s · recall 1.00
106 s · recall 0.54
58 s · recall 0.59
23 s · recall 0.58
0
100
200
300
Mean warning lead time (s), misses = 0